

PORT ARANSAS PIER, TX, USA

WMO: 994110

Lat: 27.8259N

Lon: 97.0506W

Elev: 0

StdP: 101.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	5.2	-4.5	2.6	9.0	-2.0	3.2	9.8	18.1	11.3	15.4	14.0	8.4	0	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.7	30.1	25.7	29.7	25.9	29.5	25.8	27.5	29.0	27.1	28.8	26.8	28.7	5.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.7	28.7	26.5	22.1	28.6	26.1	21.5	28.4	87.1	29.1	85.1	28.9	83.6	28.8	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.4	14.6	12.1	1.0	31.5	1.4	1.4	0.0	32.6	-0.8	33.4	-1.5	34.2	-2.5	35.2		
			-0.4	27.8	1.1	0.8	-1.2	28.3	-1.8	28.8	-2.4	29.3	-3.2	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	13.9	14.9	17.5	21.4	24.7	27.3	27.9	28.3	27.2	24.2	19.3	15.4	
	DBStd	5.98	4.11	3.70	3.25	2.39	1.65	0.99	0.82	0.84	1.80	3.12	4.16	4.30	
	HDD10.0	38	16	8	2	0	0	0	0	0	0	0	2	9	
	HDD18.3	447	141	100	50	6	0	0	0	0	0	4	38	108	
	CDD10.0	4376	137	146	235	342	457	520	556	567	517	439	283	177	
	CDD18.3	1742	3	4	25	98	199	270	298	309	267	184	68	17	
	CDH23.3	16595	1	2	5	152	1334	3021	3529	3821	3077	1497	155	2	
	CDH26.7	4376	0	1	1	9	39	718	1086	1366	965	189	1	0	
Wind	WSAvg	6.7	6.1	6.5	6.7	7.6	7.9	7.3	6.6	6.2	6.3	6.6	6.3	6.1	
Precipitation	PrecAvg	870	53	53	66	53	83	67	80	50	150	106	71	42	
	PrecMax	1269	159	135	137	219	315	225	351	183	421	357	253	153	
	PrecMin	462	0	2	2	5	0	0	0	1	33	0	0	8	
	PrecStd	241	45	44	42	52	85	56	100	46	94	99	68	39	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	21.5	23.8	28.0	28.0	29.7	30.1	30.7	31.3	29.5	26.2	23.6	
		MCWB	16.9	18.7	17.9	16.1	23.3	26.0	25.8	25.9	24.6	25.6	23.0	20.6	
	2%	DB	20.2	20.4	22.6	25.4	27.5	29.2	29.6	30.2	30.1	28.9	25.5	22.5	
		MCWB	18.4	19.0	20.2	23.0	25.0	25.7	25.8	26.0	25.6	25.4	22.7	20.9	
	5%	DB	19.6	19.8	21.8	24.7	27.2	29.0	29.4	29.8	29.7	28.3	25.0	21.8	
		MCWB	18.2	18.6	20.3	23.0	24.8	25.6	25.8	26.0	25.8	25.1	22.4	20.5	
	10%	DB	19.0	19.1	21.2	24.1	26.8	28.6	29.1	29.6	29.4	27.8	24.3	21.1	
		MCWB	17.8	18.0	20.0	22.7	24.6	25.4	25.7	26.0	25.6	24.6	21.9	19.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	20.3	22.2	25.1	26.6	27.8	27.6	28.0	27.6	26.9	24.6	22.2
			MCDB	20.2	20.6	22.5	25.2	27.0	28.8	28.9	29.3	29.4	28.3	25.9	22.8
2%		WB	19.4	19.8	21.6	24.3	26.1	27.1	27.0	27.7	26.9	26.3	23.7	21.6	
		MCDB	19.7	20.1	22.0	24.7	26.8	28.5	28.7	29.1	29.1	28.1	24.9	22.3	
5%		WB	18.9	19.2	21.0	23.7	25.7	26.6	26.8	27.2	26.7	25.7	23.0	21.0	
		MCDB	19.3	19.6	21.5	24.3	26.6	28.3	28.6	29.0	29.0	27.7	24.4	21.7	
10%		WB	18.3	18.6	20.5	23.0	25.3	26.2	26.4	26.8	26.3	25.3	22.5	20.4	
		MCDB	18.8	19.0	21.0	23.7	26.4	28.2	28.5	28.9	28.9	27.5	24.1	21.1	
Mean Daily Temperature Range	5% DB	MDBR	4.8	4.1	3.4	2.5	1.8	1.6	1.5	1.7	2.9	3.5	4.4	5.1	
		MCDBR	4.7	4.1	3.7	3.3	2.1	1.8	1.6	1.7	2.6	2.7	3.9	4.3	
		MCWBR	4.9	3.7	3.3	2.8	2.0	1.6	1.6	1.5	2.1	2.6	4.2	4.7	
	5% WB	MCDDBR	4.0	3.2	1.9	1.9	1.7	1.8	1.5	1.4	2.3	3.1	3.3	3.8	
		MCWBR	4.8	3.7	2.3	2.2	2.0	1.9	1.5	1.5	2.2	3.4	3.9	4.5	
Clear-Sky Solar Irradiance	taub	0.364	0.371	0.388	0.434	0.475	0.451	0.469	0.453	0.441	0.394	0.380	0.353		
	taud	2.484	2.467	2.414	2.290	2.207	2.341	2.306	2.351	2.345	2.473	2.464	2.530		
	Ebn at Noon	878	902	904	867	827	842	826	839	841	865	853	874		
	Edh at Noon	94	104	115	134	146	127	131	125	122	102	96	87		
All-Sky Solar Radiation	RadAvg	2.98	3.54	4.57	5.45	6.14	6.73	6.61	6.32	5.23	4.56	3.41	2.69		
	RadStd	0.33	0.58	0.48	0.52	0.41	0.35	0.55	0.34	0.44	0.46	0.36	0.36		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+81

Nomenclature: See separate page